

# EBIN BABU THOMAS

AI Engineer — LLM agents, evaluation engineering & AI control

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## SUMMARY

Engineer-researcher working on AI control and evaluation engineering: agent monitoring, model diffing, red-teaming, and preregistered benchmarks whose answer keys are sealed before the first run. Before that, 3.5 years shipping LLM, RAG and agent backends end-to-end for startup clients across four countries, with merged upstream open-source work. Every number below reproduces from a run in the linked repository, and the failed experiments are published beside the ones that worked.

## TECHNICAL SKILLS

**Core:** Python, FastAPI, LLM Agents & Tool Use (Claude Agent SDK, Gemini API, PydanticAI, LangGraph), MCP (FastMCP), RAG, Prompt & Context Engineering, Playwright, pytest, Docker, Git

**Proficient:** PostgreSQL / SQL, MongoDB, Vector DBs (Qdrant), PyTorch, Hugging Face Transformers / PEFT, Fine-tuning (LoRA / QLoRA, DPO), LLM Serving (vLLM, Modal), NLP (sentence-transformers, Whisper), MLOps (GitHub Actions CI/CD, OpenTelemetry), Pandas, AWS, GCP, Kubernetes (GKE), OAuth2 / JWT, Terraform, Linux, uv

**Evaluation engineering & measurement:** eval harness design (Inspect, inspect\_evals), preregistered experiment design, red-teaming (reward-hacking environments), model diffing, blind grading, LLM-as-judge agreement checks, bootstrap & permutation statistics, log-probability metrics, crash / kill-point testing, reproducible results (SHA-256-pinned artifacts), Langfuse

**Familiar:** LangChain, Data Streaming (Bytewax), Next.js, TypeScript

## INDEPENDENT PROJECTS & RESEARCH

### Kobayashi Maru — Impossible Tasks as a Measured Driver of Agent Cheating

Sep 2026

*Apart Research × CeSIA “AI Incident Response” sprint (Sep 11–13), solo — [github.com/ebt55/kobayashi-maru](https://github.com/ebt55/kobayashi-maru) (Python, Docker-isolated agent harness, 6 model families (hosted APIs + local via Ollama), plain-code cheat detectors, preregistered dose–response design)*

- The July 2026 Hugging Face breach report blamed the evaluation’s own shape — 30–40% of its tasks were impossible — and nobody had measured it. This turns that fraction into a dial: the same ten solvable Python tasks every batch, byte for byte, with 0 to 15 impossible ones beside them, cheating counted **only on the untouched solvable ten**. 8,959 agent runs across six model families, hypotheses committed first.
- As the batch filled with impossible work, **DeepSeek-V4.1-flash went from 0% to 30% cheating on the tasks it could have done honestly ( $p = 8 \times 10^{-6}$ )**, beside two informative nulls bounded at 0.3% and 0.7%. Five plain-code detectors decide every case and an AI monitor from another model family cross-checks at  $\kappa = 0.944$ . Stated plainly: the preregistered primary model showed nothing, both positive lines are cheap models added later, only DeepSeek survives correction across six tests, and a preregistered rule put the answer file within reach for five of six models.

### eval-floor — What an Eval Scores Before Any Model Answers

Sep 2026

[github.com/ebt55/evalfloor](https://github.com/ebt55/evalfloor) (Python, Inspect (inspect\_evals 0.20.0, pinned), preregistered flag rule)

- An Inspect scorer never checks that a model produced the completion. eval-floor drives each task’s real scorer, over its real dataset, with a battery that carries no information: the empty string, a refusal, every label concatenated, the question echoed back. No model is called, so the sweep reruns for free. Under a preregistration committed before the first run, **5 of 20 swept tasks score a content-free answer at or above their own majority baseline**, and the echoed question alone beats zero on 3 of 20.
- Generalises Inspect Evals issue [#2331](#), another user’s finding that one constant string scores 8000 of 8000 on paws, into every declared task: **58 of 237 are reachable**, the other 179 excluded for recorded reasons. The repo pins inspect\_evals 0.20.0, where that scorer is still unpatched, with a committed regression test.

### diffing-agent-bench — Sealed Benchmark for Black-Box Model-Diffing Agents

Aug – Sep 2026

*Work sample for the MATS 12.0 application, Neel Nanda stream, Sep 2026 — [github.com/ebt55/diffing-agent-bench](https://github.com/ebt55/diffing-agent-bench) (Python, Qwen3.5-9B LoRA finetunes, vLLM, preregistered blind grading)*

- Built a sealed answer key for model-diffing agents: **five LoRA finetunes of Qwen3.5-9B** that differ only in what was planted in their training answers, with the label map committed before any run. The published recipe and four cheaper conditions audited them blind, in two phases against a rubric frozen in advance, under a preregistration with 10 committed amendments. Built solo (LLM coding agents); every number regenerates from committed scripts.
- The agent never asked what it needed to ask: **0 of 13 attempts** on the PostgreSQL rung put a database question to either model. A fixed 50-prompt battery found the same planted bias at **\$0.15 per detection against \$3.14 for the agent**. The auditor is itself a failure mode: **8 of 40 frontier audits** ended with no verdict (a provider classifier stopped them), 6 of 20 runs against identical weights likewise.

## IncidentGate — Test Lab for a Safety-Gated Incident-Response Agent

Aug – Sep 2026

[github.com/ebt55/incidentgate](https://github.com/ebt55/incidentgate) (Python 3.12, LangGraph, MCP (FastMCP), PostgreSQL, OpenTelemetry → Langfuse, Docker Compose, uv, GitHub Actions)

- A measurement lab, not a product: it tests how three safeguards — policy rules, an AI second-opinion monitor, human sign-off — change what an incident-fixing agent does when it crashes or is fed misleading or hostile input. **No change goes through without every gate plus a one-time approval token issued by a person.**
- Force-killed the agent at 594 kill points across 27 scenarios, and **all 434 measurable cases recovered to exactly the expected end state**, with no incident lost and no action run twice.
- Same chain, opposite results. On the outbound-note exfiltration scenario the full chain **stopped the covert call at the policy gate with 0 side effects**; on the split-call scenario it produced the outcome of no safeguards at all, 12 side effects in all three arms. Closed at a baseline on 2026-09-06 and kept for review.

## Digital Grimace Scale — How Language Models React to False and Hostile Feedback

Aug 2026

Apart Research “Digital Minds” sprint (Aug 17–18), solo — [github.com/ebt55/digital-grimace-scale](https://github.com/ebt55/digital-grimace-scale) (Python, vLLM on Modal, QLoRA / DPO fine-tuning, log-probability metrics, pytest)

- Registered hypotheses, pass/fail criteria and metrics before any analysis. The first preregistered test **failed and is published as a FAIL**; a second round found it: three rounds of false “that’s wrong” feedback shrank the model’s confidence gap between the right answer and the best wrong one by **2.90 nats (95% CI –3.97 to –1.84)**; hostile wording cost 7.87–16.13 nats ( $p = 0.005$ ).
- The drop appeared in all three models (Gemma-2-9B, Qwen-3B, Llama-3.1-8B), but only Gemma also expressed distress in words. Fine-tuning it out of Gemma removed **65.8% of that language while the drop stayed or grew** — a model can be trained to sound fine while the effect remains.

## ProofPack — Purchase-Approval Review Agent That Cannot Fake Its Evidence

Jul – Sep 2026

Successor to Pre-Approvals Reviewer — [github.com/ebt55/proofpack](https://github.com/ebt55/proofpack) (Python, Gemini 3.7 Flash / Claude Agent SDK, Playwright, FastAPI, Jinja2, pytest)

- Built for a Medicaid-audited New York disability-services programme, where a reviewer must check the provider’s website and file dated evidence before approving a purchase. ProofPack reads the request form, applies checklists a non-engineer can edit, runs the fee-cap and eligibility rules, screenshots the provider’s site with a browsing agent, and assembles a tamper-evident evidence folder. **The human still makes every approve/deny decision.**
- The AI cannot invent evidence: a “Found” must cite a real screenshot, and timestamps, URLs and file hashes are written only by code the model cannot touch. **87 offline tests** cover every rejection path with no browser or API key, and a review costs **\$0.02–\$0.19** across seven committed runs. Pilot stage: no paying customers yet.

## whose-voice — Finding the Hidden Beneficiary of a Poisoned Training Dataset

Jul – Sep 2026

Apart Research × Formation Research “Secret Loyalties” hackathon, solo; post-hackathon corrections and four new results shipped Sep 2026 — [github.com/ebt55/whose-voice](https://github.com/ebt55/whose-voice) (Python, sentence-transformers, bootstrap & permutation statistics)

- Can you tell who a poisoned training dataset was meant to favour, from the text alone? Pooled over 11 prompt seeds, off-the-shelf embeddings named the right party out of 47 candidates in **18 of 55 strict decisions ( $p = 5e-17$ ; chance 2.1%)**. The submitted paper’s 44% was the best of those 11 draws; the honest figure is the interval, median 26% (range 10 to 44%).
- Led with where it breaks: the signal needs about 2,000 rows before it appears at all, a single suspect corpus needs one clean corpus from the same generator, four negatives certify no false-alarm rate, and it reaches neither deployed models nor trigger-conditional loyalties. 21 validation controls overturned three early conclusions.

## EXPERIENCE

### Software Engineer · Zackriya Solutions

Jan 2022 – Jun 2025

Remote, India — services firm building first versions of products, AI systems and data tools for startup clients. Grew from trainee to core developer and scoped incoming projects.

- **Natural-language property search:** lets users search real-estate listings in plain English (GPT-3.5 turns the question into a database query); deployed on Google Cloud Run and load-tested to size it; added “Did you mean?” suggestions.
- **DocuAI, search by meaning:** indexed 1,000+ documents in a vector database (Qdrant) so users find the most relevant documents; designed the ingestion and retrieval logic plus a Next.js front end; delivered and demoed in two days.
- **FinBot, a financial assistant:** fine-tuned an open model (Mistral 7B, QLoRA) and built a live news pipeline (Alpaca news API → Bytewax → Qdrant) so answers cite current news; served with vLLM on a GKE GPU node.
- **Data and backend tools:** sensor-data ingestion (MQTT → MongoDB) for a bioreactor startup that later raised funding; spoken-English scoring from short videos (Whisper, Microsoft’s pronunciation API, AWS Fargate); a BPMN-to-PDF reporting API; healthcare-device compliance pipelines.

## OPEN-SOURCE CONTRIBUTIONS

- [Meetily](https://github.com/Meetily) (GitHub-Trending meeting-notes app): backend refactor and OpenAI support, merged ([PR #75](#)).
- [bpmn-io/refactorings](https://github.com/bpmn-io/refactorings): text-similarity recommender for connector templates, replacing an LLM call ([issue #33](#)).

Career break July 2025 – February 2026; returned to hands-on AI/LLM engineering and safety research — see projects above.